CONIC SECTION

JEE EXERCISES

Exercise J-I

STANDARD & SHIFTED PARABOLA

1. The equation of the circle drawn with the focus of the parabola $(x - 1)^2 - 8y = 0$ as its centre and touching the parabola at its vertex is :

परवलय $(x - 1)^2 - 8y = 0$ की नाभि को केन्द्र मान कर बनाये गये वृत्त जोकि परवलय को उसके शीर्ष पर स्पर्श करता है, का समीकरण gS

(A) $x^2 + y^2 - 4y = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$
(C) $x^2 + y^2 - 2x - 4y = 0$ (D) $x^2 + y^2 - 2x - 4y + 1 = 0$

Ans. D

Sol. $(x - 1)^2 = 8y \Rightarrow 4a = 8 \Rightarrow a = 2$

Vertex = (1,0) focus = (1,2)

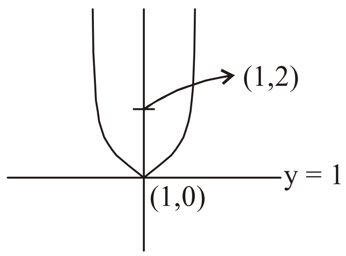
distance between focus a& vertex is r.

r = 2

Equation of circle

$(x - 1)^2 + (y - 2)^2 = 4$

$x^2 + y^2 - 2x - 4y + 1 = 0$



2. If $a \neq 0$ and the line $2bx + 3cy + 4d = 0$ passes through the points of intersection of the parabola $y^2 = 4ax$ and $x^2 = 4ay$, then :

यदि $a \neq 0$ तथा रेखा $2bx + 3cy + 4d = 0$ परवलय के प्रतिच्छेदन बिन्दु से जाती हो तो $y^2 = 4ax$ and $x^2 = 4ay$, तब :

(A) $d^2 + (3b - 2c)^2 = 0$ (B) $d^2 + (3b + 2c)^2 = 0$
(C) $d^2 + (2b - 3c)^2 = 0$ (D) $d^2 + (2b + 3c)^2 = 0$

Ans. D

Sol. Given parabolas are

$y^2 = 4ax$ ..... (i)

$x^2 = 4ay$ ..... (ii)

Putting the value of y from (ii) in (i), we get

$\frac{x^2}{16a^2} = 4ax \Rightarrow x(x^3 - 64a^3) = 0 \Rightarrow x = 0, 4a$

from (ii), y = 0, 4a. Let $A \equiv (0, 0)$; $B \equiv (4a, 4a)$

Since, given line $2bx + 3cy + 4d = 0$ passes through A and B,

$\therefore$ d = 0 and 8ab + 12ac = 0

$\Rightarrow$ $2b + 3c = 0$. ($\because a \neq 0$)

Obviously, $d^2 + (2b + 3c)^2 = 0$

PARAMETRIC COORDINATES OF PARABOLA

3. PN is an ordinate of the parabola $y^2 = 4ax$ (P on $y^2 = 4ax$ and N on x-axis). A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q. NQ meets the tangent at the vertex A in apoint T such that AT = kNP, then the value of k is : (where A is the vertex)

परवलय $y^2=4ax$ की कोटि PN है जहां P परवलय पर स्थित है। अक्ष के समान्तर एक सरल रेखा NP को समद्विभाजित करती है तथा वक्र को Q पर मिलती है। NQ शीर्ष A पर खींची गयी स्पर्श रेखा को बिन्दु T पर इस प्रकार मिलती है कि AT = kNP हो, तो k का मान है : (जहां A शीर्ष है)

(A) 3/2 (B) 2/3 (C) 1 (D) 1/3

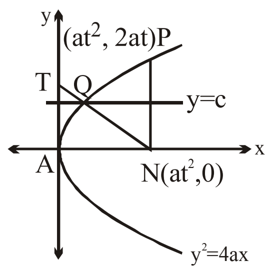
Ans. B

Sol. Equation of PN : $x = at^2$

$y = c$ bisects PN

$\therefore \quad c = at$

which cuts the parabola at Q



$\Rightarrow \quad c^2 = 4ax \qquad \therefore \qquad x = \dfrac{c^2}{4a}$

$\therefore \quad Q\left(\dfrac{c^2}{4a}, c\right) = Q\left(\dfrac{at^2}{4}, at\right)$

$\therefore \quad$ Equation of NQ: $y - 0 = \dfrac{at - 0}{\dfrac{at^2}{4} - at^2}(x - at^2)$

$y = \dfrac{-4}{3t}(x - at^2)$

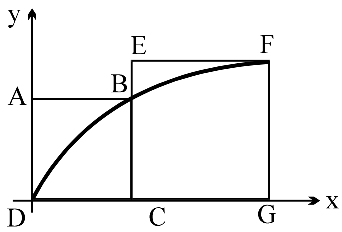
which cuts x = 0 at $\left(0, \dfrac{4at}{3}\right)$

$\therefore \quad T = \dfrac{4at}{3}$ and NP = 2at

$\therefore \quad \dfrac{AT}{NP} = \dfrac{4at/3}{2at} = \dfrac{2}{3}$ Ans.

4. ABCD and EFGC are squares and the curve $y = k\sqrt{x}$ passes through the origin D and the points B and F. The ratio $\dfrac{FG}{BC}$ is :

चित्र में ABCD और EFGC वर्ग है और वक्र $y = k\sqrt{x}$ मूल बिन्दु D और बिन्दुओं B और F से गुजरता है तब $\frac{FG}{BC}$ है :



(A) $\frac{\sqrt{5}+1}{2}$ (B) $\frac{\sqrt{3}+1}{2}$ (C) $\frac{\sqrt{5}+1}{4}$ (D) $\frac{\sqrt{3}+1}{4}$

Ans. A

Sol. $y^2 = k^2x \Rightarrow y^2 = 4ax$ where $k^2 = 4a$

$B = \left(at_1^2, 2at_1\right);\ F = \left(at_2^2, 2at_2\right) \quad \{t_1 > 0, t_2 > 0\}$

to find $\frac{FG}{BC} = \frac{2at_2}{2at_1} = \frac{t_2}{t_1}$

now DC = BC $\Rightarrow at_1^2 = 2at_1 \Rightarrow t_1 = 2$

also $at_2^2 - at_1^2 = 2at_2$

$t_2^2 - 4 = 2t_2$

$t_2^2 - 2t_2 - 4 = 0$

$t_2 = \frac{2 \pm \sqrt{4+16}}{2}$ but $t_2 > 0 \quad \therefore \quad t_2 \neq \frac{2-\sqrt{20}}{2}$

$t_2 = \left(\sqrt{5}+1\right)$

$\therefore \quad \frac{t_2}{t_1} = \frac{\sqrt{5}+1}{2}$ Ans.

5. Let P be the point (1, 0) and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is :
माना परवलय $y^2 = 8x$ पर बिन्दु Q है तथा कोई बिन्दु P(1, 0) तब PQ के मध्य बिन्दु का बिन्दुपथ होगा :
(A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x + 2 = 0$ (C) $x^2 + 4y + 2 = 0$ (D) $x^2 - 4y + 2 = 0$

Ans. A

Sol. $2h = \frac{\alpha^2}{8} + 1$

$8(2h-1) = \alpha^2$ $\qquad\qquad 2k = \alpha$

$8(2h-1) = 4k^2$

$4h - 2 = k^2$

$y^2 - 4x + 2 = 0$

6. The vertex A of the parabola $y^2 = 4ax$ is joined to any point P on it and PQ is drawn at right angles to AP to meet the axis in Q. Projection of PQ on the axis is equal to :

(A) twice the length of latus rectum (B) the length of latus rectum

(C) half the length of latus rectum (D) one fourth of the length of latus rectum

परवलय $y^2 = 4ax$ के शीर्ष A को परवलय पर स्थित बिन्दु P से मिलने पर यह PQ पर समकोण बनाता है जबकि Q अक्ष पर है तब PQ का अक्ष पर प्रक्षेप होगा :

(A) नाभिलम्ब का दुगुना (B) नाभिलम्ब की लम्बाई के बराबर

(C) नाभिलम्ब की लम्बाई का आधा (D) नाभिलम्ब की लम्बाई का एक चौथाई

Ans. B

Sol. hyperbola $\frac{x^2}{16} - \frac{y^2}{48} = 1$

7. The graph of the equation $x + y = x^3 + y^3$ is the union of :

(A) line and an ellipse (B) line and a parabola

(C) line and hyperbola (D) line and a point

समीकरण $x + y = x^3 + y^3$ निम्न वक्रों का संयुक्त समीकरण है :

(A) रेखा तथा दीर्घवृत्त (B) रेखा तथा परवलय

(C) रेखा तथा अतिपरवलय (D) रेखा तथा बिन्दु

Ans. A

Sol. Given $(x + y) = (x + y)(x^2 - xy + y^2) \Rightarrow (x + y)(x^2 - xy + y^2 - 1) = 0$

$\therefore$ Either $x + y = 0$ or $x^2 - xy + y^2 - 1 = 0$

Now $x + y = 0$ represent a line.

Also $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = \begin{vmatrix} 1 & \frac{-1}{2} & 0 \\ \frac{-1}{2} & 1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = \frac{-3}{4} \neq 0$ and $h = \frac{-1}{2}$, $a = 1 = b$, so $h^2 < ab$

$\therefore x^2 - xy + y^2 - 1 = 0$ represent an ellipse .

Hence locus of $P(x, y)$ satisfying above equation is the union of line and an ellipse.

STANDARD & SHIFTED ELLIPSE

8. A focus of an ellipse is at the origin. The directrix is the line x = 4 and the eccentricity is $\frac{1}{2}$. Then the length of the semi-major axis is :

एक दीर्घवृत्त की एक नाभि मूल बिन्दु पर है। रेखा x = 4 उसकी नियता है तथा उसकी उत्केन्द्रता $\frac{1}{2}$ है। तो उसके अर्ध–दीर्घ अक्ष की yEckbZ gS

(A) $\frac{2}{3}$ (B) $\frac{4}{3}$ (C) $\frac{5}{3}$ (D) $\frac{8}{3}$

Ans. D

Sol. $e = \frac{5}{2} \Rightarrow \frac{a}{e} - ae = 4 \Rightarrow a\left[2-\frac{1}{2}\right] = 4 \Rightarrow a \cdot \frac{3}{2} = 4$

$a = \frac{8}{3} \Rightarrow a\left[\frac{1}{2}-2\right] = 4$

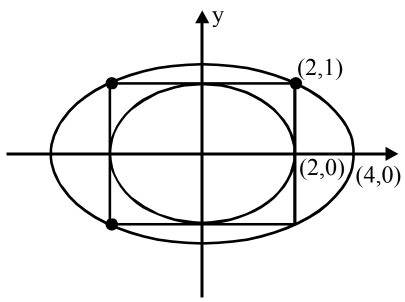
9. The ellipse $x^2 + 4y^2 = 4$ is inscribed in a rectangle aligned with the coordinate axes, which in turn is inscribed in another ellipse that passes through the point (4, 0). Then the equation of the ellipse is : **[AIEEE-2009]**

दीर्घवृत $x^2 + 4y^2 = 4$ निर्देशांक अक्षों से सरेखित एक आयत के अन्तर्गत है जो स्वयं बिन्दु (4, 0) से जाने वाले दूसरे दीर्घवृत के अन्तर्गत gSrc bl nh?kZor dk lehdj.k gS **[AIEEE-2009]**

(A) $4x^2 + 48y^2 = 48$ (B) $4x^2 + 64y^2 = 48$ (C) $x^2 + 16y^2 = 16$ (D) $x^2 + 12y^2 = 16$

Ans. D

Sol.



Ellipse $x^2 + 4y^2 = 4$ (Given)

Eq^n. of the ellipse (required)

$$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$$

Ellipse passes through (2, 1)

therefore $\frac{4}{16}+\frac{1}{b^2}=1 \Rightarrow b^2=\frac{4}{3}$

$$\frac{x^2}{16}+\frac{y^2}{4/3}=1 \Rightarrow \frac{x^2}{16}+\frac{3y^2}{4}=1$$

$\Rightarrow x^2 + 12y^2 = 16$

10. An ellipse is drawn by taking a diameter of the circle $(x-1)^2 + y^2 = 1$ as its semi-minor axis and a diameter of the circle $x^2 + (y-2)^2 = 4$ as its semi-major axis. If the centre of the ellipse is at the origin and its axes are the coordinate axes, then the equation of the ellipse is : **[AIEEE-2012]**

वृत $(x-1)^2 + y^2 = 1$ के एक व्यास को अर्ध लघु अक्ष लेकर तथा वृत $x^2 + (y-2)^2 = 4$ के एक व्यास को अर्ध दीर्घ अक्ष लेकर एक nh?kZo`r [khapk x;kA ;fn nh?kZo`r dk dsUnz ewy fcanq j gS rFkk bl ds v{k funsZ'kkad v{k gSa rks nh?kZo`r dk lehdj.k gS **[AIEEE-2012]**

(A) $x^2 + 4y^2 = 16$ (B) $4x^2 + y^2 = 4$ (C) $x^2 + 4y^2 = 8$ (D) $4x^2 + y^2 = 8$

Ans. A

Sol. Let the equation of ellipse be

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

from the given conditions

$a = 4$ and $b = 2$

$\therefore$ Eq of ellipse is

$$\frac{x^2}{16}+\frac{y^2}{4}=1$$

or $x^2 + 4y^2 = 16$

STANDARD & SHIFTED HYPERBOLA

11. For the hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, which of the following remains constant α varies?

(A) Abscissae of vertices (B) Abscissae of foci

(C) Eccentricity (D) Directrix

अतिपरवलय $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, के लिये α का मान परिवर्तित करने पर निम्न में से कौन अचर रहेगा?

(A) शीर्षों के भुज (B) नाभियों के भुज

(C) उत्केन्द्रता (D) नियता

Ans. B

Sol. The given equation of hyperbola is

$$\frac{x^2}{\cos^2\alpha} - \frac{y^2}{\sin^2\alpha} = 1$$

$\Rightarrow$ $a = \cos\alpha$, $b = \sin\alpha$

$\Rightarrow$ $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \tan^2\alpha} = \sec\alpha$

$\Rightarrow$ $ae = 1$

$\therefore$ foci $(\pm 1, 0)$

$\therefore$ foci remain constant with respect to α.

12. The foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide. Then the value of b^2 is :

दीर्घवृत्त $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ तथा अतिपरवलय $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ की नाभियां सम्पाती हों तो b^2 का मान होगा :

(A) 9 (B) 1 (C) 5 (D) 7

Ans. D

Sol. For ellipse $a^2 = 16$ $\Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{\sqrt{16 - b^2}}{4}$ $\Rightarrow$ focii $\equiv (\pm ae, 0) = \left(\pm\sqrt{16 - b^2}, 0\right)$

For hyperbola, $e = \sqrt{1 + \frac{b^2}{a^2}} = \frac{5}{4}$ $\therefore$ focii $\equiv (\pm ae, 0) = (\pm 3, 0)$

$\therefore$ $\sqrt{16 - b^2} = 3$ $\Rightarrow$ $b^2 = 7$

13. In eccentricity of conjugate hyperbola of the given hyperbola $\left|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}\right| = 3$ is e', then value of 8e' is :

दिये गये अतिपरवलय $\left|\sqrt{(x-1)^2 + (y-2)^2} - \sqrt{(x-5)^2 + (y-5)^2}\right| = 3$ के संयुग्मी अतिपरवलय की उत्केन्द्रता e' है, तो 8e' का मान होगा :

(A) 12 (B) 14 (C) 17 (D) 10

Ans. D

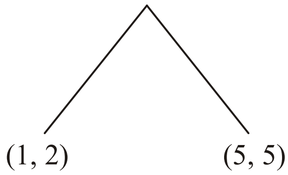
Sol. $(PS_1 - PS_2) = 2a$

$2a = 3 \Rightarrow a = \frac{3}{2}$

$2ae = 5$

$2\times\frac{3}{2}\times e = 5$

$e = \frac{5}{3}$



$\frac{1}{e^2}+\frac{1}{e'^2}=1$

$\frac{9}{25}+\frac{1}{e'^2}=1 \Rightarrow \left(\frac{1}{e'}\right)^2=\frac{16}{25} \Rightarrow e' = \frac{5}{4}$

$8e' = 8\times\frac{5}{4} = 10$

14. A hyperbola, having the transverse axis of length $2\sin\theta$, is confocal with the ellipse $3x^2 + 4y^2 = 12$. Then its equation is :

एक अतिपरवलय, जिसकी अनुप्रस्थ अक्ष की लम्बाई $2\sin\theta$ है तथा दीर्घवृत्त $3x^2 + 4y^2 = 12$ की नाभियाँ सम्पाती हैं। तब अतिपरवलय d hl elhdj.kgS

(A) $x^2\,\mathrm{cosec}^2\theta - y^2\sec^2\theta = 1$ (B) $x^2\sec^2\theta - y^2\,\mathrm{cosec}^2\theta = 1$

(C) $x^2\sin^2\theta - y^2\cos^2\theta = 1$ (D) $x^2\cos^2\theta - y^2\sin^2\theta = 1$

Ans. A

Sol. Given $3x^2 + 4y^2 = 12$ an ellipse

$\therefore\ a^2 = 4\ \ b^2 = 3$

$\therefore\ e = \sqrt{1-\frac{3}{4}} \Rightarrow e = \frac{1}{2}$

$\therefore$ It's focus will be $(\pm1, 0)$

Since hyperbola is confocal to given ellipse, therefore $\pm ae = \pm1$, but $a = \sin\theta$ given

$\therefore\ e = \frac{1}{\sin\theta}$, Now $b^2 = a^2(e^2 - 1)$

$\beta^2 = \sin^2\theta\,\frac{\cos^2\theta}{\sin^2\theta} \Rightarrow b^2 = \cos^2\theta$

Hence required equation will be,

$\frac{x^2}{\sin^2\theta} - \frac{y^2}{\cos^2\theta} = 1$

$\Rightarrow\ x^2\mathrm{cosec}^2\theta - y^2\sec^2\theta = 1$

15. Let a and b be non-zero real numbers. Then, the equation $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ represents :

(A) four straight lines, when $c = 0$ and a, b are of the same sign.

(B) two straight lines and a circle, when $a = b$, and c is of sign opposite to that of a.

(C) two straight lines and a hyperbola, when a and b are of the same sign and c is of sign opposite to that of a

(D) a circle and an ellipse, when a and b are of the same sign and c is of sign opposite to that of a.

माना a एवं b अशून्य वास्तविक संख्याएं है। समीकरण $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$ प्रदर्शित करता है :

(A) चार सरल रेखाओं को जबकि $c = 0$ तथा a, b समान चिन्ह के हो

(B) दो सरल रेखाओं तथा एक वृत को यदि $a = b$, तथा c का चिन्ह a के विपरीत है।

(C) दो सरल रेखाओं तथा एक अतिपरवलय को जहाँ पर a एवं b समान चिन्ह के हो तथा c, a के विपरीत चिन्ह का है

(D) एक वृत तथा एक दीर्घवृत को जहाँ पर, a एवं b समान चिन्ह के है तथा c, a के विपरीत चिन्ह का है

Ans. B

Sol. $(ax^2 + by^2 + c)(x^2 - 5xy + 6y^2) = 0$

either $x^2 - 5xy + 6y^2 = 0 \Rightarrow$ two straight lines passing through origin.

or $ax^2 + by^2 + c = 0$

(A) If c = 0, and a and b are of same sign then it will represent a point.

(B) If a = b, c is of sign opposite to a then it will represent circle.

(C) When a & b are of same sign and c is of sign opposite to a then it will represent ellipse.

(D) This is clearly incorrect.

16. Consider a branch of the hyperbola, $x^2 - 2y^2 - 2\sqrt{2}\,x - 4\sqrt{2}\,y - 6 = 0$ with vertex at the point A. Let B be one of the end points of its latus rectum. If C is the focus of the hyperbola nearest to the point A, then the area of the triangle ABC is :

माना बिन्दु A पर स्थित शीर्ष वाला अतिपरवलय $x^2 - 2y^2 - 2\sqrt{2}\,x - 4\sqrt{2}\,y - 6 = 0$ है। माना इसके नाभिलम्ब के शीर्षो में से एक ' kh"kZB है। यदि बिन्दु A के निकटतम अतिपरवलय की नाभि C है, तब ΔABC का क्षेत्रफल है :

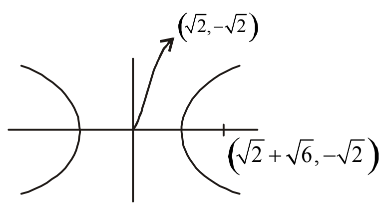
(A) $1-\sqrt{\frac{2}{3}}$ (B) $\sqrt{\frac{3}{2}}-1$ (C) $1+\sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{3}{2}}+1$

Ans. B

Sol. The given equation is

$(x - \sqrt{2})^2 - 2(y + \sqrt{2})^2 = 4$

$\frac{(x-\sqrt{2})^2}{4} - \frac{(y+\sqrt{2})^2}{2} = 1$



$a = 2, \quad b = \sqrt{2}$

hence eccentricity $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{\frac{3}{2}}$

$$\text{Area} = \frac{1}{2}a(e-1) \times \frac{b^2}{a}$$

$$= \left(\sqrt{\frac{3}{2}} - 1\right) \text{ sq. units.}$$

TANGENT & NORMAL

17. The straight line joining any point P on the parabola $y^2 = 4ax$ to the vertex and perpendicular from the focus to the tangent at P, intersect at R, then the equation of the locus of R is :

वक्र $y^2 = 4ax$ के किसी बिन्दु P और शीर्ष को मिलाने वाली रेखा और बिन्दु P पर स्पर्श रेखा पर नाभि से लम्ब बिन्दु R पर प्रतिच्छेद करती g$r c fcUnqR का बिन्दु पथ होगा :

(A) $x^2 + 2y^2 - ax = 0$ (B) $2x^2 + y^2 - 2ax = 0$ (C) $2x^2 + 2y^2 - ay = 0$ (D) $2x^2 + y^2 - 2ay = 0$

Ans. B

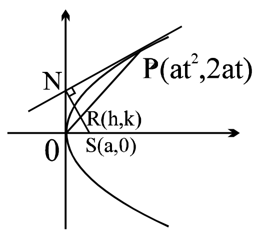
Sol. T : $ty = x + at^2$ ....(1)

line perpendicular to (1) through (a,0)

$tx + y = ta$ ....(2)

equation of OP : $y - \frac{2}{t}x = 0$ ....(3)

from (2) & (3) eleminating t we get locus



18. The normal chord of a parabola $y^2 = 4ax$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is :

परवलय $y^2 = 4ax$ की किसी बिन्दु, जिसके भुज और कोटि बराबर है, पर अभिलम्ब जीवा का नाभि पर अन्तरित कोण है :

(A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$

Ans. D

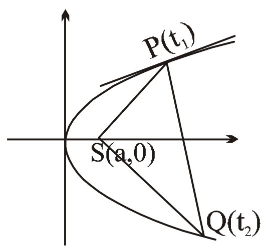
Sol. $a t_1^2 = 2at_1 \Rightarrow t_1 = 2$; P(4a , 4a)

$$t_2 = -t_1 - \frac{2}{t_1} = -3$$

$\therefore$ Q (9a ,– 6a)

$$\therefore \quad m_{SP} = \frac{4a}{4a - a} = \frac{4}{3}$$

$$\left.m_{SQ} = \frac{-6a}{9a - a} = -\frac{3}{4}\right] \angle PSQ = 90^\circ$$



19. PQ is a normal chord of the parabola $y^2 = 4ax$ at P, A being the vertex of the parabola. Through P a line is drawn parallel to AQ meeting the x–axis in R. Then the length of AR is :

(A) equal to the length of the latus rectum

(B) equal to the focal distance of the point P

(C) equal to twice the focal distance of the point P

(D) equal to the distance of the point P from the directrix.

चफ परवलय $y^2 = 4ax$ की P पर अभिलम्ब जीवा है यदि P से एक रेखा AQ के समान्तर खींची जाती है जो x अक्ष को R पर मिलती है t gkaA परवलय का शीर्ष है, तब AR की लम्बाई :

(A) नाभिलम्ब की लम्बाई के बराबर होगी
(B) बिन्दु P की नाभि दूरी के बराबर होगी
(C) बिन्दु P की नाभि दूरी की दुगुनी होगी
(D) नियता से बिन्दु P की दूरी के बराबर होगी

Ans. C

Sol. $t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_1t_2 + t_1^2 = -2$

Equation of the line through P parallel to AQ

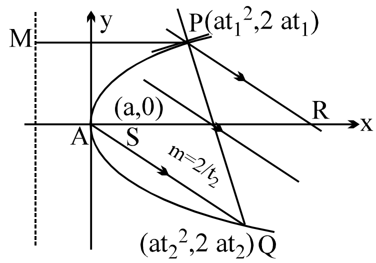
$$y - 2at_1 = \frac{2}{t_2}(x - at_1^2)$$

put $y = 0 \Rightarrow x = at_1^2 - at_1t_2$

$= at_1^2 - a(-2 - t_1^2)$

$= 2a + 2at_1^2 = 2(a + at_1^2)$

= twice the focal distance of P



20. Let BC be the latus rectum of the parabola $y^2 = 4x$ with vertex A. Minimum length of the projection of BC on a tangent drawn in the portion BAC is :

शीर्ष A वाले परवलय $y^2 = 4x$ की नाभिलम्ब BC है, तो BAC के क्षेत्र में किसी बिन्दु पर खीचें गये स्पर्श रेखा पर BC के प्रक्षेप (Projection) की न्यूनतम लम्बाई होगी :

(A) 2 (B) $2\sqrt{3}$ (C) $2\sqrt{2}$ (D) $2+\sqrt{2}$

Ans. C

Sol. Projection of BC on the tangent = | BC sin θ |

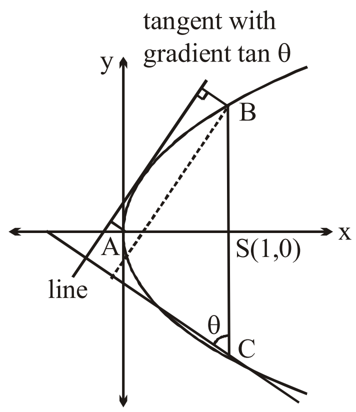
$$= 4\,|\sin\theta|,\ \theta \in \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$$

at A, $\theta \to 90° \Rightarrow$ projection = 4

at B, $\theta \to 45°$ or $135°$

Hence $(BC)_{min.} = 4 \sin 45°$

$$= \frac{4}{\sqrt{2}} = 2\sqrt{2} \text{ Ans.}$$



21. The normal at the point $(bt_1^2, 2bt_1)$ on a parabola meets the parabola again in the point $\left(bt_2^2, 2bt_2\right)$, then :

परवलय के बिन्दु $(bt_1^2, 2bt_1)$ पर खींचा गया अभिलम्ब पुनः परवलय के बिन्दु $(bt_2^2, 2bt_2)$ पर मिलता है तब :

(A) $t_2 = t_1 + \frac{2}{t_1}$ (B) $t_2 = -t_1 - \frac{2}{t_1}$ (C) $t_2 = -t_1 + \frac{2}{t_1}$ (D) $t_2 = t_1 - \frac{2}{t_1}$

Ans. B

Sol. It is a fundamental theorem.

22. The tangent and normal at P(t), for all real positive t, to the parabola $y^2 = 4ax$ meet the axis of the parabola in T and G respectively, then the angle at which the tangent at P to the parabola is inclined to the tangent at P to the circle passing through the points P, T and G is :

यदि t के सभी वास्तविक धनात्मक मानों के लिए पवलय $y^2 = 4ax$ के बिन्दु P(t) से स्पर्श रेखा और अभिलम्ब परवलय की अक्ष को क्रमशः T और G पर मिलते हें तब P पर परवलय की स्पर्श रेखा और P पर वृत्त की स्पर्श रेखा के मध्य का कोण जहां वृत्त बिन्दुओं P, T और G से गुजरती है, है :

(A) $\cot^{-1}t$ (B) $\cot^{-1}t^2$ (C) $\tan^{-1}t$ (D) $\tan^{-1}t^2$

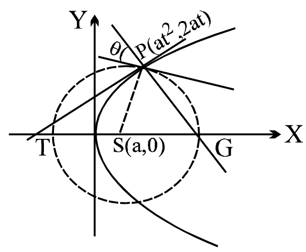
Ans. C

Sol. slope of tangent $= \frac{1}{t}$ (m_1) at P on parabola

$$\text{slope of PS} = \frac{2at}{a(t^2-1)} = \frac{2t}{t^2-1} \text{(normal to the circle)}$$

$\therefore$ $$\text{slope of tangent at P on circle} = \frac{1-t^2}{2t}(m_2)$$

$\therefore$ $$\tan\theta = \frac{\frac{1}{t} - \frac{1-t^2}{2t}}{1 + \frac{1-t^2}{2t^2}} = \frac{\left(2-1+t^2\right)2t^2}{2t(1+t^2)} = t$$



$\therefore$ $\theta = \tan^{-1}t \Rightarrow$ (C)

23. Equation of the other normal to the parabola $y^2 = 4x$ which passes through the intersection of those at $(4, -4)$ and $(9a, -6a)$ is :

परवलय $y^2 = 4x$ के बिन्दुओं $(4, -4)$ तथा $(9a, -6a)$ से खीचें गये अभिलम्बो के प्रतिच्छेद बिन्दु से खीचे गये अभिलम्ब का समीकरण gksxk:

(A) $5x - y + 115 = 0$ (B) $5x + y - 135 = 0$ (C) $5x - y - 115 = 0$ (D) $5x + y + 115 = 0$

Ans. B

Sol. Equation of normal at (4, – 4) is $y = 2x - 12$ and equation of normal at (9, – 6) is y = 3x – 33a. Point of intersection of both these normals is (21, 30)

Now $y = mx - 2m - m^3$

$30 = 21m - 2m - m^3 = 19m - m^3$

$\Rightarrow \quad m^3 - 19m + 30 = 0$

$m = 2, 3, -5$

24. Let P be the point on the parabola, $y^2 = 8x$ which is at a minimum distance from the centre C of the circle, $x^2 + (y + 6)^2 = 1$. Then the equation of the circle, passing through C and having its centre at P is : **[JEE(Main)2016]**

माना परवलय $y^2 = 8x$ का P का ऐसा बिन्दु है जो वृत्त $x^2 + (y + 6)^2 = 1$, के केन्द्र C से न्यूनतम दूरी पर है, तो उस वृत्त का समीकरण C से होकर जाता है तथा जिसका केन्द्र P पर है, है : **[JEE(Main)2016]**

(A) $x^2 + y^2 - \frac{x}{4} + 2y - 24 = 0$ (B) $x^2 + y^2 - 4x + 9y + 18 = 0$

(C) $x^2 + y^2 - 4x + 8y + 12 = 0$ (D) $x^2 + y^2 - x + 4y - 12 = 0$

Ans. C

Sol. $P(2t^2, 4t)$

Normal at P passes through (0, –6)

$tx + y = 4t + 2t^3$

$-6 = 4t + 2t^3$

$t^3 + 2t + 3 = 0$

$t = -1$

$P(2, -4)$

25. If the tangent at (1, 7) to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ then the value of c is : **[JEE(Main)2018]**

यदि वक्र $x^2 = y - 6$ के बिन्दु (1, 7) पर बनी स्पर्शरेखा वृत $x^2 + y^2 + 16x + 12y + c = 0$ को स्पर्श करती है, तो c का मान है : **[JEE(Main)2018]**

(A) 85 (B) 95 (C) 195 (D) 185

Ans. B

Sol. $x^2 = y - 6$

$\left(\frac{dy}{dx}\right)_{(1,7)} = 2$

$T\,(1, 7) = 2x - y + 5 = 0$

distance of center from tangent = radius

$$r = \left|\frac{-16+6+5}{\sqrt{5}}\right| = \sqrt{5}$$

$$\sqrt{g^2+f^2-c} = \sqrt{5}$$

$\Rightarrow c = 95$

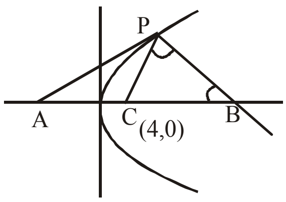
26. Tangent and normal are drawn at P(16, 16) on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A ανδ B, ρεσπεχτιϖελψ ΙφΧ ισ τηε χεντρε οφ τηε χιρχλε τηρουγη τηε ποιντσ Π, A ανδ B ανδ $\angle CPB = \theta$, then a value of $\tan\theta$ is : **[JEE(Main)2018]**

परवलय $y^2 = 16x$ के एक बिन्दु P(16, 16) पर स्पर्श रेखा तथा अभिलम्ब खींचे जाते है तो परवलय के अक्ष को बिन्दुओं क्रमशः A तथा B पर प्रतिच्छेद करते हैं। यदि बिन्दुओं P, A तथा B से होकर जाने वाले वृत का केन्द्र C है तथा $\angle CPB = \theta$ तो $\tan\theta$ का एक मान है: **[JEE(Main)2018]**

(A) 3 (B) $\frac{4}{3}$ (C) $\frac{1}{2}$ (D) 2

Ans. D

Sol.



$$\left(-\frac{dx}{dy}\right) = -t = -2$$

Slope of normal at P = $-\tan\theta = -2$

$\tan\theta = 2$

27. The point on the ellipse $x^2 + 2y^2 = 6$, whose distance from the line $x + y = 7$ is minimum is :

(A) (2, 3) (B) (2, 1) (C) (1, 0) (D) None of these

दीर्घवृत $x^2 + 2y^2 = 6$ पर बिन्दु जिसकी रेखा $x + y = 7$ से दूरी न्यूनतम है :

(A) (2, 3) (B) (2, 1) (C) (1, 0) (D) इनमें से कोई नहीं

Ans. B

Sol. E : $\frac{x^2}{6}+\frac{y^2}{3}=1$ $a^2 = 6$ $\Rightarrow a = \sqrt{6}$

$b^2 = 3$ $\Rightarrow b = \sqrt{3}$

Point on ellipse $\left(\sqrt{6}\cos\theta, \sqrt{3}\sin\theta\right)$

distance from line $x + y = 7$

$$d = \frac{\left(\sqrt{6}\cos\theta + \sqrt{3}\sin\theta - 7\right)}{\sqrt{2}}$$

$$\left(a\cos\theta + b\sin\theta\right) \in \left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$$

$$d_{min.} = \frac{\left|[-3,3]-7\right|}{\sqrt{2}} = \frac{\left|[-10,-4]\right|}{\sqrt{2}}$$

$d_{min.} = 2\sqrt{2} \leftarrow$ correct but we don't have to find this.

Equation of tangent at $P\left(\sqrt{6}\cos\theta, \sqrt{3}\sin\theta\right)$

$$\frac{x\sqrt{6}\cos\theta}{6} + \frac{y\sqrt{3}\sin\theta}{3} = 1 \qquad ....(1)$$

it is parallel to $x + y = 7$ so slopes equal

$$\Rightarrow -\frac{\cos\theta}{\sqrt{6}} \times \frac{\sqrt{3}}{\sin\theta} = -1$$

$$\tan\theta = \frac{1}{\sqrt{2}} \left\langle \begin{array}{l} \sin\theta = \frac{1}{\sqrt{3}} \\ \cos\theta = \frac{\sqrt{2}}{\sqrt{3}} \end{array} \right.$$

$$\therefore P\left(\sqrt{6} \times \frac{\sqrt{2}}{\sqrt{3}}, \sqrt{3} \times \frac{1}{\sqrt{3}}\right) \Rightarrow P(2,1)$$

28. The locus of the foot of perpendicular drawn from the centre of the ellipse $x^2 + 3y^2 = 6$ on any tangent to it is : **[JEE(Main)2014]**

दीर्घवृत $x^2 + 3y^2 = 6$ के केंद्र से इसकी किसी स्पर्श रेखा पर खींचे गए लंब के पाद का बिंदु पथ है : **[JEE(Main)2014]**

(A) $(x^2 + y^2)^2 = 6x^2 + 2y^2$
(B) $(x^2 + y^2)^2 = 6x^2 - 2y^2$
(C) $(x^2 - y^2)^2 = 6x^2 + 2y^2$
(D) $(x^2 - y^2)^2 = 6x^2 - 2y^2$

Ans. A

Sol. $K = mh \pm \sqrt{6m^2 + 2}$ __________(1)

$\left(\dfrac{K}{h}\right)m = -1$ __________(2)

Eliminating m from (1) and (2)

Locus is

$(x^2 + y^2)^2 = 6x^2 + 2y^2$

29. The locus of a point $P(\alpha, \beta)$ moving under the condition that the line $y = \alpha x + \beta$ is a tangent to the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ is :

(A) an ellipse (B) a circle (C) a parabola (D) a hyperbola

बिन्दु $P(\alpha, \beta)$, जो इस प्रतिबन्ध के अन्तर्गत घूम रहा है, कि $y = \alpha x + \beta$ अतिपरवलय $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ की स्पर्श रेखा है, का fcUnqपथ :

(A) एक दीर्घवृत्त है (B) एक वृत्त है (C) एक परवलय है (D) एक अतिपरवलय है

Ans. D

Sol. On comparing $y = mx + \sqrt{a^2m^2 - b^2}$ and $y = \alpha x + \beta$

\ $\alpha = m, \beta = \sqrt{a^2m^2 - b^2}$

$\Rightarrow \beta^2 = a^2\alpha^2 - b^2$

$\Rightarrow a^2\alpha^2 - \beta^2 = b^2$

$\Rightarrow$ Hyperbola

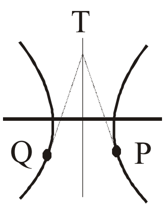
30. Tangent are drawn to the hyperbola $4x^2 - y^2 = 36$ at the points P and Q. If these tangents intersect at the point T(0, 3) then the area (in sq. units) of ΔPTQ is : **[JEE(Main)2018]**

एक अतिपरवलय $4x^2 - y^2 = 36$ के बिंदुओं P तथा Q पर स्पर्श रेखाएँ खींची जाती है। यदि यह स्पर्शरेखाएँ बिंदु T(0, 3) पर काटती g$r ks$\Delta PTQ$ का क्षेत्रफल (वर्ग इकाइयों में) है : **[JEE(Main)2018]**

(A) $60\sqrt{3}$ (B) $36\sqrt{5}$ (C) $45\sqrt{5}$ (D) $54\sqrt{3}$

Ans. C

Sol.



Eqaution of tangent at P

$$\frac{x\sec\theta}{3}-\frac{y\tan\theta}{6}=1$$

it will pass though (0, 3)

$$0-\frac{3\tan\theta}{6}=1$$

$\tan\theta = -2 \Rightarrow \sec\theta = \sqrt{5}$

$$\text{area} = \frac{1}{2}\left|6\sec\theta\right|\left(|6\tan\theta|+3\right)$$

$= 45\sqrt{5}$

31. Let P(6, 3) be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the normal at the point P intersects the x-axis at (9, 0), then the eccentricity of the hyperbola is : **[JEE-2011]**

माना अतिपरवलय $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ पर बिन्दु P(6, 3) स्थित है। यदि P पर अभिलम्ब x-अक्ष को (9, 0) पर काटता है तो अतिपरवलय की mRl fUi;r kgksxh: **[JEE-2011]**

(A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

Ans. B

Sol. Equation of normal at P(6, 3) on $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $\frac{a^2x}{6}+\frac{b^2y}{3}=a^2e^2$

It intersects x-axis at (9, 0)

$\Rightarrow \quad a^2\frac{9}{6}=a^2e^2 \quad \Rightarrow \quad e=\sqrt{\frac{3}{2}}$

COMMON TANGENTS & COMMON NORMAL

32. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to :

वक्रों $y^2 = x - 1$ और $x^2 = y - 1$ के मध्य की न्यूनतम दूरी है :

(A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{5\sqrt{2}}{4}$ (C) $\frac{7\sqrt{2}}{4}$ (D) $\frac{\sqrt{2}}{4}$

Ans. A

Sol. (A) $\left(\frac{x}{3}\right)^2+\left(\frac{y}{4}\right)^2=\cos^2 t+\sin^2 t=1 \Rightarrow$ An ellipse

(B) $x^2-2=-2\cos t=-2\left(2\cos^2\frac{t}{2}-1\right) \Rightarrow$ A parabola

(C) $\sqrt{x}=\tan x$, $\sqrt{y}=\sec t \quad (x\geq 0, y\geq 0)$

$\sec^2 t-\tan^2 t=1 \quad \Rightarrow \quad y-x=1 \Rightarrow$ a line tangent

(D) $x^2=1-\sin t=2-(1+\sin t)=2-\left(\sin\frac{t}{2}+\cos\frac{t}{2}\right)^2=2-y^2 \Rightarrow$ a circle

33. Consider two curves C_1: $(y-\sqrt{3})^2=4(x-\sqrt{2})$ and C_2: $x^2+y^2=(6+2\sqrt{2})x+2\sqrt{3}y-6(1+\sqrt{2})$, then :

(A) C_1 and C_2 touch each other only at one point.
(B) C_1 and C_2 touch each other exactly at two points.
(C) C_1 and C_2 intersect (but do not touch) at exactly two points.
(D) C_1 and C_2 neither intersect nor touch each other.

माना C_1: $(y-\sqrt{3})^2=4(x-\sqrt{2})$ और C_2: $x^2+y^2=(6+2\sqrt{2})x+2\sqrt{3}y-6(1+\sqrt{2})$ दो वक्र है, तब :

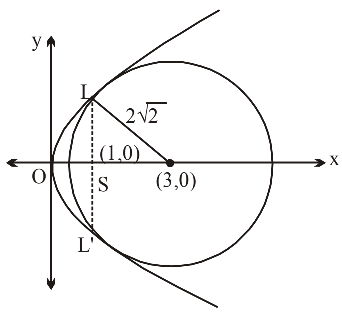
(A) ह$_1$ और C_2 एक दूसरे को एक बिन्दु पर स्पर्श करेंगे
(B) ह$_1$ और C_2 एक दूसरे को दो बिन्दुओं पर स्पर्श करेंगे
(C) ह$_1$ और C_2 दो बिन्दुओं पर प्रतिच्छेद करेंगे। (स्पर्श नहीं)
(D) ह$_1$ और C_2 न तो प्रतिच्छेद करते हैं नहीं स्पर्श

Ans. B

Sol. Given $C_1:(y-\sqrt{3})^2=4(x-\sqrt{2})$ .....(1)

and $C_2:\left(x-(3+\sqrt{2})\right)^2+(y-\sqrt{3})^2=\left(3+\sqrt{2}\right)^2+3-6\left(1+\sqrt{2}\right)$

$\left[\left(x-\sqrt{2}\right)-3\right]^2+\left(y-\sqrt{3}\right)^2=8$ .....(2)



Let $y-\sqrt{3}=Y$ and $x-\sqrt{2}=X$

$C_1'=Y^2=4X$ .....(3)

$C_2'=(X-3)^2+Y^2=8$....(4)

Solving $(X-3)^2+4X=8$

$X^2-2X+1=0$

$(X-1)^2=0$

$X=1$

Hence the two curves touch each other exactly at two points.

34. Equation of the common tangent to the ellipses, $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$, is :

निम्न में से कौनसी सरल रेखा दोनों वक्रों $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$ तथा $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$ की उभयनिष्ठ स्पर्श रेखा है :

(A) $ay = bx + \sqrt{a^4 - a^2b^2 + b^4}$ (B) $by = ax - \sqrt{a^4 + a^2b^2 + b^4}$

(C) $ay = bx - \sqrt{a^4 + a^2b^2 + b^4}$ (D) $by = ax + \sqrt{a^4 - a^2b^2 + b^4}$

Ans. B

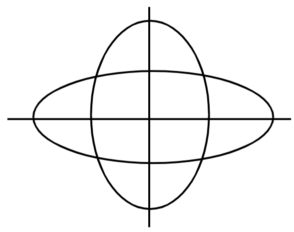
Sol. Equation of a tangent to $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$

$$y = mx \pm\sqrt{(a^2+b^2)m^2+b^2} \qquad \text{....(1)}$$

If (1) is also a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$ then

$(a^2 + b^2)m^2 + b^2 = a^2m^2 + a^2 + b^2$ (using $c^2 = a^2m^2 + b^2$)

$b^2m^2 = a^2 \Rightarrow m^2 = \frac{a^2}{b^2} \Rightarrow m = \pm\frac{a}{b}$



$$y = \pm\frac{a}{b}x \pm \sqrt{(a^2+b^2)\frac{a^2}{b^2}+b^2}$$

$$by = \pm ax \pm \sqrt{a^4+a^2b^2+b^4}$$

Note : Although there can be four common tangents but only one of these appears in B

35. Two common tangents to the circle $x^2 + y^2 = 2a^2$ and parabola $y^2 = 8ax$ are :

वृत्त $x^2 + y^2 = 2a^2$ तथा परवलय $y^2 = 8ax$ की दो उभयनिष्ठ स्पर्श रेखा है होगी :

(A) $x = \pm (y + 2a)$ (B) $y = \pm (x + 2a)$ (C) $x = \pm (y + a)$ (D) $y = \pm (x + a)$

Ans. B

Sol. $C_1 : x^2 + y^2 = 2a^2$

Equation of tangent

$$y = mx \pm \sqrt{2a^2m^2+2a^2} \qquad \text{\_\_\_\_\_\_\_(1)}$$

$P : y^2 = 8ax$

$y = mx \pm \frac{2a}{m}$ ________(2)

from (1) & (2)

$\frac{4a^2}{m^2} = 2a^2m^2 + 2a^2$

$4a^2 = 2a^2m^4 + 2a^2m^2$

$2 = m^4 + m^2$

$m^4 + m^2 - 2 = 0$

$m^4 + 2m^2 - m^2 - 2 = 0$

$m^2 (m^2 + 2) - 1(m^2 + 2) = 0$

$m = \pm 1$

$y = \pm(x + 2a) \rightarrow$ Equation of common tangents

ELLIPSE : MISCELLANEOUS

36. Two sets A and B are as under : $A = \{(a, b) \in R \times R : |a - 5| < 1$ and $|b - 5| < 1\} : B\{(a, b) \in R \times R : 4(a - 6)^2 + 9(b - 5)^2 \le 36\}$. Then : **[JEE(Main)2018]**

(A) $A \cap B = \phi$ (an empty set) (B) neither $A \subset B$ not $B \subset A$

(C) $B \subset A$ (D) $A \subset B$

दो समुच्चय A तथा B निम्न प्रकार के है – $A = \{(a, b) \in R \times R : |a - 5| < 1$ तथा $|b - 5| < 1\} : B \{(a, b) \in R \times R : 4(a - 6)^2 + 9(b - 5)^2 \le 36\}$. तो : **[JEE(Main)2018]**

(A) $A \cap B = \phi$ (एक रिक्त समुच्चय) (B) न तो $A \subset B$ और न ही $B \subset A$

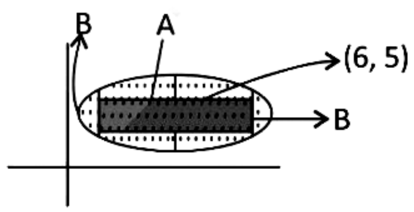
(C) $B \subset A$ (D) $A \subset B$

Ans. D

Sol. Let $a - 6 = x$ $b - 5 = y$

$|x + 1| < 1$ $|y| < 1$ $4x^2 + 9y^2 \le 36$

$-2 < x < 0$ $-1 < y < 1$



Boundaries of square are lying in ellipse

$\Rightarrow A \subset B$

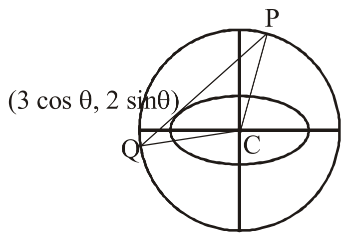
37. A tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with centre C meets its director circle at P and Q. Then the product of the slopes of CP and CQ, is :

दीर्घवृत्त $\frac{x^2}{9}+\frac{y^2}{4}=1$, जिसका केन्द्र C है, पर स्पर्श रेखा नियामक वृत्त को P तथा Q पर काटती है तो CP तथा CQ के ढालों का xqku gksxk:

(A) $\frac{9}{4}$ (B) $\frac{-4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$

Ans. B

Sol. The equation of the tangent at $(3\cos\theta, 2\sin\theta)$ on $\frac{x^2}{9}+\frac{y^2}{4}=1$ is



$$\frac{x}{3}\cos\theta+\frac{y}{2}\sin\theta=1 \qquad \text{... (i)}$$

The equation of the director circle is

$$x^2+y^2=9+3=13 \qquad \text{... (ii)}$$

The combined equation of CP and CQ is obtained by homogenising equation (ii) with (i). Thus combined equation is

$$x^2+y^2=13\left(\frac{x}{3}\cos\theta+\frac{y}{2}\sin\theta\right)^2$$

$$\Rightarrow \quad \left(\frac{13}{9}\cos^2\theta-1\right)x^2+\frac{13}{3}\sin\theta\cos\theta\,xy+\left(\frac{13}{4}\sin^2\theta-1\right)y^2=0$$

$\therefore$ Product of the slopes of CP and CQ

$$\frac{\text{coefficient of } x^2}{\text{coefficient of } y^2}=\frac{\frac{13}{9}\cos^2\theta-1}{\frac{13}{4}\sin^2\theta-1}=\frac{13\cos^2\theta-9}{13\sin^2\theta-4}\times\frac{4}{9}=\frac{13\cos^2\theta-9}{13-13\cos^2\theta-4}\times\frac{4}{9}=-\frac{4}{9}$$

[Note: $m_1+m_2=\frac{-2h}{b}$; $m_1m_2=\frac{a}{b}$]

38. The normal at a variable point P on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e' such that :

दीर्घवृत्त $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ पर स्थित किसी बिन्दु पर खींची गयी अभिलम्ब अक्षों को क्रमशः बिन्दु Q तथा R पर मिलती है तो QR के e/; fcUnq dk fcUnqiFk dh mRdsUnzrk dk eku e' इस प्रकार है कि :

(A) e' is independent of e
(B) $e' = 1$
(C) $e' = e$
(D) $e' = 1/e$

Ans. C

Sol. Normal at point P $(a\cos\theta, b\sin\theta)$

$$\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2 \qquad ...(1)$$

it meets axes of ellipse at

$$Q=\left(\frac{\left(a^2-b^2\right)\cos\theta}{a},0\right)$$

$$R=\left(0,-\frac{\left(a^2-b^2\right)\sin\theta}{a}\right)$$

mid point of Q & R is M (h, k).

$$2h=\frac{\left(a^2-b^2\right)\cos\theta}{a}$$

$$\Rightarrow \cos\theta=\frac{2ab}{a^2-b^2} \qquad ...(2)$$

$$2k=-\frac{\left(a^2-b^2\right)\sin\theta}{b}$$

$$\Rightarrow \sin\theta=-\frac{2bk}{\left(a^2-b^2\right)} \qquad ...(3)$$

from [Equation (2)]^2 + [equation (3)]^2

$$\frac{x^2}{\frac{\left(a^2-b^2\right)^2}{4a^2}}+\frac{y^2}{\frac{\left(a^2-b^2\right)^2}{4b^2}}=1$$

$$e'^2 = 1 - \frac{b^2}{a^2} = e^2 \Rightarrow e' = e$$

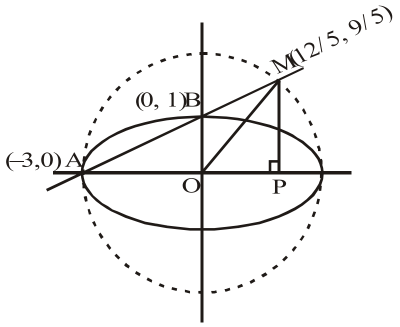
39. The line passing through the extremity A of the major axis and extremity B of the minor axis of the ellipse $x^2 + 9y^2 = 9$ meets its auxiliary circle at the point M. Then the area of the triangle with vertices at A, M and the origin O is : **[JEE-2009]**

दीर्घवृत्त $x^2 + 9y^2 = 9$ के दीर्घ–अक्ष के सिरा A तथा लघु–अक्ष के सिरा B से जाने वाली रेखा सहायत–वृत्त को बिन्दु M पर काटती है। ' kh"kZ cUnq ks A, M तथा मूलबिन्दु O से बनने वाले त्रिभुज का क्षेत्रफल होगा : **[JEE-2009]**

(A) $\frac{31}{10}$ (B) $\frac{29}{10}$ (C) $\frac{21}{10}$ (D) $\frac{27}{10}$

Ans. D

Sol. Area of triangle AOM = $\frac{1}{2}$ AO. PM



$\Rightarrow$ Eqaution of AM is $y = \frac{1}{3}(x+3)$

$x - 3y + 3 = 0$ which is chord of auxiliary circle $x^2 + y^2 = 9$, and PM is ordinate of point M

$\Rightarrow (3y-3)^2 + y^2 = 9 \Rightarrow y = \frac{9}{5} = PM$ $\Rightarrow$ Area of triangle $= \frac{1}{2} \cdot 3 \cdot \frac{9}{5} = \frac{27}{10}$.

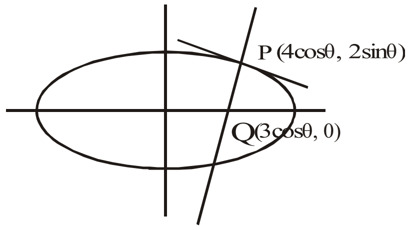
40. The normal at a point P on the ellipse $x^2 + 4y^2 = 16$ meets the x-axis at Q. If M is the mid point of the line segment PQ, then the locus of M intersects the latus rectums of the given ellipse at the point : **[JEE-2009]**

दीर्घवृत्त $x^2 + 4y^2 = 16$ के किसी बिन्दु P पर अभिलम्ब x-अक्ष से बिन्दु Q पर मिलता है। यदि रेखाखण्ड PQ का मध्य बिन्दु M है, r c M का बिन्दुपथ दिए गए दीर्घवृत्त के नाभिलम्बों को किन बिन्दुओं पर काटता है : **[JEE-2009]**

(A) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{2}{7}\right)$ (B) $\left(\pm\frac{3\sqrt{5}}{2}, \pm\frac{\sqrt{19}}{4}\right)$ (C) $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$ (D) $\left(\pm 2\sqrt{3}, \pm\frac{4\sqrt{3}}{7}\right)$

Ans. C

Sol. $\frac{x^2}{16}+\frac{y^2}{4}=1$



$4x\sec\theta - 2y\,\text{cosec}\theta = 12$

$x = 3\cos\theta$

$Q \equiv (3\cos\theta, 0)$

$2\eta = 7\ \chi o\sigma\theta$

$2k = 2\sin\theta$

$\frac{4x^2}{49}+\frac{y^2}{1}=1$ ...(i)

L.R $\Rightarrow x = 2\sqrt{3}$

Putting in (i)

$y = \pm\frac{1}{7}$ $\therefore$ $\left(\pm 2\sqrt{3}, \pm\frac{1}{7}\right)$

AUXILLIARY CIRCLE & ECCENTERIC ANGLE OF ELLIPSE AND HYPERBOLA

41. If α & β are the eccentric angles of the extremities of a focal chord of an standard ellipse, then the eccentricity of the ellipse is :

यदि α तथा β मानक दीर्घवृत्त के नाभि जीवा के सिरों के उत्केन्द्रिय कोण हे तो दीर्घवृत्त की उत्केन्द्रियता का मान होगा :

(A) $\frac{\cos\alpha + \cos\beta}{\cos(\alpha+\beta)}$ (B) $\frac{\sin\alpha - \sin\beta}{\sin(\alpha-\beta)}$ (C) $\frac{\cos\alpha - \cos\beta}{\cos(\alpha-\beta)}$ (D) $\frac{\sin\alpha + \sin\beta}{\sin(\alpha+\beta)}$

Ans. D

Sol. $\frac{x}{a}\cos\frac{\alpha+\beta}{2}+\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$; $\frac{ae}{a}\cos\frac{\alpha+\beta}{2}=\cos\frac{\alpha-\beta}{2}$

$$\Rightarrow \quad e = \frac{\cos\frac{\alpha-\beta}{2}}{\cos\frac{\alpha+\beta}{2}}\cdot\frac{2\sin\frac{\alpha+\beta}{2}}{2\sin\frac{\alpha+\beta}{2}} = \frac{\sin\alpha + \sin\beta}{\sin(\alpha+\beta)}$$

42. If P is any point on ellipse with foci S_1 & S_2 and eccentricity is $\frac{1}{2}$ such that $\angle PS_1S_2 = \alpha$, $\angle PS_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, then $\cot\frac{\alpha}{2}$, $\cot\frac{\gamma}{2}$, $\cot\frac{\beta}{2}$ are in :

यदि P दीर्घवृत्त पर कोई बिन्दु है जिसकी नाभियां S_1 तथा S_2 है तथा उत्केन्द्रिता $\frac{1}{2}$ इस प्रकार है कि $\angle PS_1S_2=\alpha$, $\angle PS_2S_1 = \beta$, $\angle S_1PS_2 = \gamma$, तो $\cot\frac{\alpha}{2}$, $\cot\frac{\gamma}{2}$, $\cot\frac{\beta}{2}$ निम्न श्रेणी में है :

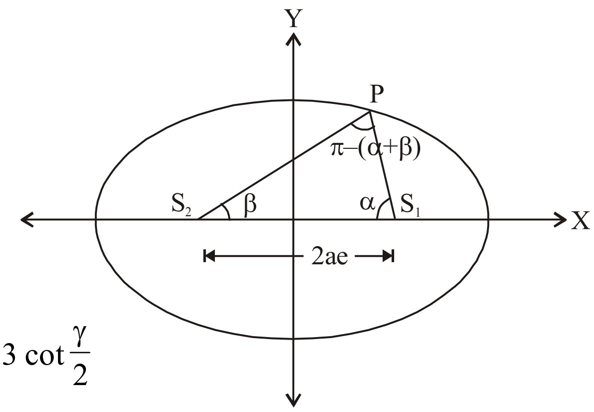
(A) A.P. (B) G.P. (C) H.P. (D) *NOT* A.P., G.P. & H.P.

Ans. A

Sol. By sine rule in ΔPS_1S_2, we get $\dfrac{2ae}{\sin(\alpha+\beta)} = \dfrac{S_1P}{\sin\beta} = \dfrac{S_2P}{\sin\alpha} = \dfrac{2a}{\sin\alpha+\sin\beta}$

$$\Rightarrow e = \frac{\sin(\alpha+\beta)}{\sin\alpha+\sin\beta} \Rightarrow \frac{e}{1} = \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}$$

Now $\dfrac{1-e}{1+e} = \tan\dfrac{\alpha}{2}\tan\dfrac{\beta}{2} = \dfrac{1-\frac{1}{2}}{1+\frac{1}{2}} = \dfrac{\frac{1}{2}}{\frac{3}{2}} = \dfrac{1}{3}$



$\therefore \tan\dfrac{\alpha}{2}\tan\dfrac{\beta}{2} = \dfrac{1}{3}$ ... (1)

Also we know that

$$\cot\frac{\alpha}{2} + \cot\frac{\beta}{2} + \cot\frac{\gamma}{2} = \cot\frac{\alpha}{2}\cot\frac{\beta}{2}\cot\frac{\gamma}{2} = 3\cot\frac{\gamma}{2}$$

$\Rightarrow 2\cot\dfrac{\gamma}{2} = \cot\dfrac{\alpha}{2} + \cot\dfrac{\beta}{2} \Rightarrow \cot\dfrac{\alpha}{2}, \cot\dfrac{\gamma}{2}, \cot\dfrac{\beta}{2}$ are in A.P.

RECTANGULAR HYPERBOLA

43. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is :

आयताकार अतिपरवलय $xy = c^2$ के समान्तर जीवाओं (जिनके ढाल m है) के निकाय के मध्य बिन्दु का बिन्दुपथ होगा :

(A) $y + mx = 0$ (B) $y - mx = 0$ (C) $my - x = 0$ (D) $my + x = 0$

Ans. A

Sol. equation of chord with mid point (h, k) is $\dfrac{x}{h} + \dfrac{y}{k} = 2$; $m = -\dfrac{k}{h} \Rightarrow y + mx = 0$

44. The normal to curve $xy = 4$ at the point (1, 4) meets the curve again at :

वक्र $xy = 4$ के बिन्दु (1, 4) पर अभिलम्ब वक्र को पुनः किस बिन्दु पर मिलेगा :

(A) $(-4, -1)$ (B) $\left(-8, -\dfrac{1}{2}\right)$ (C) $\left(-16, -\dfrac{1}{4}\right)$ (D) $(-1, -4)$

Ans. C

Sol. $xy = 4$, $c^2 = 4 \Rightarrow c = 2$

point $\left(ct, \frac{c}{t}\right) = (1, 4) \Rightarrow t = \frac{1}{2}$

$m_N = t^2 = \frac{1}{4}$

Equation of normal

$(y-4) = \frac{1}{4}(x-1)$

$4y - 4 \times 4 = x - 1$

$x - 4y + 15 = 0$ ...(1)

$xy = 4$ ...(2)

from equation (1) & (2)

$x - 4 \times \frac{4}{x} + 15 = 0$

$x^2 + 15x - 16 = 0$

$x^2 + 16x - x - 16 = 0$

$x(x + 16) - 1(x + 16) = 0$

$x = -16, 1$ $1 \rightarrow$ Not possible

$x = -16, \ y = -\frac{1}{4}$

Point $= \left(-16, -\frac{1}{4}\right)$

45. Locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is a/an :

(A) ellipse (B) circle (C) hyperbola (D) parabola

उस बिन्दु का बिन्दुपथ, जिसकी वृत $x^2 + y^2 = 4$ के सापेक्ष स्पर्श जीवा अतिपरवलय $xy = 1$ की स्पर्श रेखा है, होगा :

(A) दीर्घवृत (B) वृत (C) अतिपरवलय (D) परवलय

Ans. C

Sol. Let P(h, k) be point $\rightarrow$ C.O.C. of circle

$x^2 + y^2 = 4$

$\Rightarrow 4x + ky = 4$ ...(1) $\Rightarrow y = \dfrac{4 - bx}{k}$

Equation (1) is tangent to xy = 1.

$$x \cdot \left(\frac{4 - hx}{k}\right) = 1$$

$4x - hx^2 - k = 0 \Rightarrow hx^2 - 4x + k = 0$

COT D = 0

$16 - 4hk = 0$

$4hk = 16$

$hk = 4$

$\Rightarrow xy = 4.$ its a R.H.B.

46. If the tangent and normal at a point on rectangular hyperbola cut-off intercept a_1, a_2 on x-axis and b_1, b_2 on the y-axis, then $a_1a_2 + b_1b_2$ is equal to :

यदि आयतीय अतिपरवलय के किसी बिन्दु पर स्पर्श रेखा तथा अभिलम्ब द्वारा काटे गये अन्तःखण्ड x- अक्ष पर क्रमशः a_1 एवं a_2 तथा y-अक्ष पर b_1 एवं b_2 है, तो $a_1a_2 + b_1b_2$ का मान है :

(A) 2 (B) $\dfrac{1}{2}$ (C) 0 (D) –1

Ans. C

Sol. Equation of tangent at $\left(ct, \dfrac{c}{t}\right)$

$$\frac{x}{t} + yt = 2c \qquad ...(1)$$

Equation of normal at $\left(ct, \dfrac{c}{t}\right)$

$$\left(y - \frac{c}{t}\right) = t^2(x - ct) \qquad ...(2)$$

Put y = 0 in euqation (1) & (2)

$$a_1 = 2ct,\ a_2 = ct - \frac{c}{t^3}$$

put $x = 0$ in equation (1) & (2)

$$b_1 = \frac{2c}{t},\ b_2 = \frac{c}{t} - ct^3$$

$$a_1a_2 + b_1b_2 = 0$$